DAY CARE SURGERY

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MODULE OBJECTIVE

At the end of this module, students will be able to provide day-care anesthesia for various types of surgical procedures

MODULE COMPETENCES

Provide Anesthesia for day-care surgical patients in a compassionate respectful and caring manner

LEARNING OUTCOMES

In order to achieve core competencies, students at the end of this Module will be able to:

Describe the principles and considerations of Anesthesia for day-care surgery

Describe the pharmacologic and non-pharmacologic optimization strategies of day-care surgical patients

Order and interpret appropriate diagnosing investigation for day-care anesthesia and surgery and provide intraoperative anesthesia care for various types of Day-care surgery.

Manage pain using multimodal analgesia technique for day-care surgical patients

Provide post-anesthetic care for day-care surgical patients

apply discharging criteria for day-care anesthesia

WHAT IS DAY CARE ANESTHESIA ?

Day care surgery is defined as surgery for which a patient is admitted and discharged on the same day. There has been a dramatic growth in day surgery over the past two decades.

Patient should be awake within 1.5 hours post operatively for discharge .

INTRODUCTION

The development of sophisticated medical and surgical technology and advances in anesthesia pharmacology and drug delivery system has contributed greatly to the explosive growth of Day care surgery.

Ambulatory anesthesia is administered with the dual goals of rapidly and safely establishing satisfactory condition for the performance of therapeutic or diagnostic procedures while ensuring rapid, predictable recovery with minimal postoperative squeal

WHY DAY CARE ?

Patient preference, esp. children and the elderly

Lack of dependence on the availability of hospital beds

Greater flexibility in scheduling operations

Lower incidence of infection

Lower incidence of respiratory complications

Higher volume of patients

Shorter surgical waiting lists

Lower overall procedural costs

Less preoperative testing and postoperative medications.

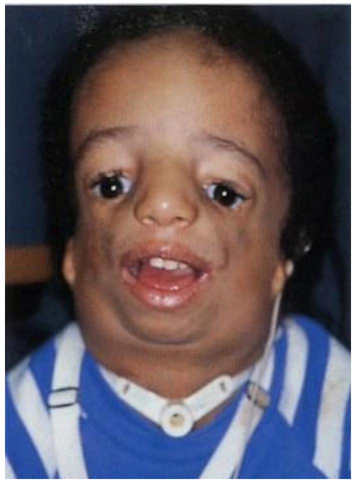
WHY DAY CARE ?

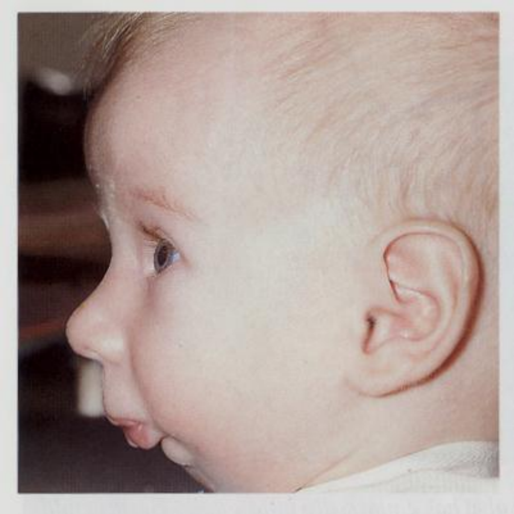
Costs of the hospital settings increased.

The incidence of death and major morbidity associated directly with day surgery is extremely low, and there are some significant advantages when compared with inpatient surgery, including reductions in staffing costs, shorter waiting times and fewer hospital acquired infections.

Plans emerged where procedures could be done apart from hospital activity.

CAN WE SCHEDULE THE FOLLOWING CASES FOR DAY CARE ?





BACK GROUND

Back ground history of daycare surgery(Nicolas 1889-1909)

RM Walters 1916-opened down town clinic

1984-society for ambulatory anesthesia.

1985-34% Of all the surgeries performed on day care bases

20th-century time when day care surgery become viable(50%of surgeries)

FACILITY

Free standing clinic

Surgical centers

In a separate fully contained units in the hospital

CRITERIA'S FOR LICENSING

Should relay on general hospital for back up care if need arises

ADVANTAGES

Protects from hospital acquired infection

Avoids family unit disruption

Minimizes lasting psychological support

Reimbursement is easily forth coming

Accepted by ensures (Accountable)

After care at home more congenital

For surgeons schedule runs in time

Nursing care acceptable

Anesthesia care is appropriate

Decreases medical costs

DISADVANTAGE

Disadvantages

- Surgical and anesthetic complications resulting in unplanned readmissions to the hospital
- Need for higher expertise level
- Possible chances of negligence in preoperative anesthetic assessment

Lesser compliance to preoperative fasting instructions and preoperative medications

- Higher anxiety levels among patients

SELECTION CRITERIA'S

Patient characteristics

Planed surgical procedure

Patient factors

PATIENT SELECTION

It is now accepted that most patients are suitable for day surgery unless there is a specific reason why an overnight stay would be required.

Social factors: The patient must understand the procedure and consent to have it performed as a day surgery. Patients who undergo general anesthesia should be looked

after by a responsible adult for the next 24 hours, and their home circumstances should be appropriate for postoperative care

Surgical factors

It is important that any procedure considered for day surgery should not carry a significant risk of major complications, nor should it require prolonged specialist care or observation.

Medical factors

Patients require a preoperative assessment to determine their physical health prior to day surgery. There has been a move away from arbitrary criteria, such as those laid down by the American Society of Anesthesiologists (ASA) or body mass index (BMI),

and it is generally felt that patients with a chronic but stable illness are better managed at home as this results in less disruption to their routine.

ESSENTIALS FOR PATIENT CARE

Physical status ASAI/ASAII

Complete history / lab investigation /signed consent /pack of information's

Any contraindications to anesthesia(24 hrs prior)

Where and when to report

How long the stay will be

How to conduct themselves personally

ESSENTIAL DIRECTIONS FOR PATIENTS

Time and place of registration

Calling in upper respiratory tract(cancellation)

NPO hrs 10 hours

Escort to and from the facility

Time and manner of discharge

Personal belongings

No driving alcohol or major decisions post operatively how and where to report complication

Cancellation(full stomach acute URTI)

PREOPERATIVE VISIT OF ANESTHETIST

A- review of records

B-standard questions asked & data recorded (questioners before day care surgery patients; adults/pediatrics &geriatrics)--

C- state of health

D-physical characteristics noted

E-explaining anesthetic procedures

GOALS OF PREOPERATIVE VISIT

Asses appropriateness of patient and procedure for ambulatory surgery

Obtain medical information needed to plan anesthetic care .

Asses factors that affect risk of anesthesia

Asses factors that might alter planned anesthetic technique

Obtain informed consent

Asses patients social situation in relation to day care surgery

Provide preoperative education to patients(NPO and medical education)

Acquaint patient with ambulatory surgery facility

Provide patient with clear expectations for anesthesia care and post operative course

ABSOLUTE AND RELATIVE CONTRAINDICATIONS DEPENDING UPON THE TYPE OF FACILITY

Procedure associated with significant blood loss and severe post operative pain

Acute concurrent illness

Poorly compensated and incompletely evaluated systemic disease (severe diabetes /asthma /coronary artery disease/

Severe systemic disease requiring intensive care follow up(unstable ASA 3&4)

Sickle cell disease

Coagulopathy

Acute substance abuse

Premature infants with significant bronchopulmonary dysplasia

Susceptibility to malignant hyperthermia

History of sudden infant death syndrome

Lack of social support

ASA CLASSIFICATIONS

ASA 1-normal healthy patient

ASA 2- patient with mild systemic disease

ASA 3-ptient with severe systemic disease that limits activity but is not incapacitating

ASA 4- a patient with an incapacitating systemic disease that is a constant threat to life

ASA 5-a moribund patient not expected to survive more than 24hrs with or with out operation

INTERVIEW

Age

Scheduled procedure

Medications

Allergies

Cigarette alcohol and drug history

Anesthetic history

Prior surgical procedure

Family history of any anesthetic problems

Social history

Birth and developmental history

Obstetrics history

Medical problems

Review of systems

Exercise tolerance

History of any air way problems

Patient concerns preferences or expectations regarding anesthesia

QUESTIONS

What operations are you having?

Do you have any medical problems?

Do you ever had a problem with your heart ,chest pain ,palpitation heart attack?

How much physical activity can you endure?

Do you get shortness of breath during normal activities?

Do you have any problems with your B/P?

Do you smoke or drink alcohol?

Do you use any chemicals?

Have you ever had bronchitis/pneumonia/asthma?

Do you have dentures/eye glasses?

Any history of OSA?

Do you have any neurological problems?

Do you have a cough?

Any history of jaundice?

History of gastritis

History of any kidney problem?

History of blood clotting problem/sickle cell disease/hemoglobin problem?

Have you received blood transfusion/do you accept blood transfusion?

Have you ever had an operation?

Any history of problem with previous anesthesia?

Any history of allergy

Is there any thing else about your health that you think I should know? and to menstruating females ask obstetrical history

PREOPERATIVE EVALUATION

History

Physical examination

Laboratory and ancillary testing

PHYSICAL EXAMINATION

Height & weight

Base line mental status

Vital signs

Air way evaluation

Evaluation of heart and lungs

Skin conditions(turgor /jaundice /pallor

Land mark for regional technique

Neurologic function

Vascular access

Extrimities (clubbing /edema /pulses

AIRWAY EVALUATION

Mallampati classification (ability to view posterior pharynx)

Mentum to thyriod distance

Mandible to hyoid bone distance

Oral opening

Nares

Quality of dentition(teeth gums dentures)

Intra oral structures ((tonsils /uvula /palates)

Mask fit (facial anatomy /beard

Range of motion of neck

obesity

DIFFICULT AIRWAY

1-Asses the likelihood and clinical impact of basic management problems

difficult intubation

difficult ventilation

difficulty with patient cooperation

2-Consider the relative merits and feasibility of basic management choices

3- develop primary and alternative strategies

A-awake intubation choices

LABORATORY INVESTIGATIONS

Hematologic and biochemical testing (LFT/RFT)

Pregnancy testing

HIV screening

ECG(Echocardiography)

Chest X/Ray

Pulmonary function test

Cervical spine X/Ray

CT scan /MRI

ADVANTAGES AND DISADVANTAGES OF PREOPERATIVE EVALUATION

Advantages

Comprehensive evaluation is possible

Ready accessibility of consultations and laboratory tests

Patients questions can be answered directly

Can obtain informed consent

Can give prescription for preoperative medications

DISADVANTAGE

Patient may still not meet with the anesthetist who may conduct anesthesia if it is practiced on group bases

Requires an extra trip to the center

Time consuming for the patient

Usually not reimbursed by the third party

May not decrease patient anxiety

TELEPHONE INTERVIEW

Advantage

Saves the patients an extra trip to the hospital

Patient has the opportunity to speak to the anesthetist

Patients questions can be answered directly disadvantages

Time consuming ‘often frustrating for the health professional

Low contact rate

Physical examination is not possible

BASIC STANDARDS FOR PRE ANESTHETIC CARE

These applies to all patients who receive anesthesia or monitored anesthesia care

STANDARDS

Anesthetists must be responsible for determining the medical status of the patient

Shall develop a plan of anesthesia care

Acquaint the patient or the responsible adult with the proposed plan

TO DEVELOP APPROPRIATE PLAN OF ANESTHESIA

A-Review the medical record

B-Interview and examine the patient

1-Discuss the medical history /previous anesthetic experience and drug therapy

2-Asses those aspects of physical condition that might affect decisions regarding peri operative risk and management

C-obtain and review tests and consultations necessary to the conduct of anesthesia

D-determine the appropriate prescription of preoperative medications as necessary to the conduct of anesthesia.

INDICATIONS FOR CARDIOLOGIST OR INTERNIST CONSULTATIONS

Recent myocardial infarction (less than 6 months)

Unstable angina

Uncontrolled severe hypertension /diabetes/COPD/significant renal or liver disease

Exercise intolerance with out an obvious cause

Significant and symptomatic cardiac arrhythmias

Recent abnormal ECG changes suggestive of coronary artery disease

Congestive cardiac failure

Bleeding disorders (hemophilia)

Hemoglobinopathies (sickle cell disease

OTHER CONSIDERATIONS

Medical problems (optimal control)

Elderly patients(physiologic not chronologic age)

Post conception age (gestation +post natal)

NPO status (with reasons)

Inappropriate candidates criteria must be individualized)

INAPPROPRIATE CANDIDATES

infants at risk

1-Premature birth

Apeunistic episodes

Failure to thrive

RDS

Bronco pulmonary dysplasia

SIDS

Uncontrolled seizure

MHT

2-morbidely obese patients

3-ASA physical status III&IV (presence of infection)

4-un cooperative patient

5- No responsible person at home

Age

Premature infants < 46 weeks of postconceptional age are at increased risk and are not an ideal candidate for Ambulatory surgery.

Anemia is a significant independent risk factor , particularly for infants less than 43 weeks of post conception age.

ASA physical status

ASA status I, II and medically stable ASA status III

The risk of complication can be reduced if pre-existing medical conditions are under good control for at least 3 months before operation.

Elderly outpatients may experience a higher incidence of peri operative event and slow recovery of fine motor skills and cognitive functions

PROCEDURES

Depends on patient characteristics

Experience of anesthetist

Resources and facilities available

Duration of surgery

Major intervention to thorax abdomen and cranial vault are not for daycare surgery

Emergency cases are not for day care surgery

Operations associated with major bleeding are not for day care surgery.

PEDIATRICS DAY CARE SURGERY

There is no single ideal anesthetic technique for day care surgery in children . However, the technique chosen should ensure minimum unpleasantness and morbidity compatible with safety for patients and good operating conditions for the surgeon. Recovery should be rapid to allow early discharge

Children are excellent candidates for day care management as they are usually healthy and predominantly require minor or intermediate surgery of short duration.

A successful pediatric day-case service is one which minimizes postoperative morbidity, has low in-patient admission rates and demonstrate high parental and child satisfaction.

Good quality anesthesia is essential to achieving these goals with experienced clinicians working in child-friendly facilities.

Important concerns

1- distinguish which child should not have anesthesia on day care basis

2-grouping of special clinical problems

respiratory status

pre op screening

traditional barriers

3- physical status

4-history (asthma /sickle cell anemia /malignancy/ croup /downs syndrome/renal/diabetes /URTI(Anesthetic implication)

5- NPO status (written instruction)

RECENT STUDIES

96% of pediatrics exhibit gastric PH of 2.5or less.

60-80%residual volume of 0.4ml /kg

Calmer children tend to have high PH

ASPIRATION PROPHYLAXIS

Gastro kinetics

H2 receptor antagonists

Non particulate anti acids

ADVANTAGES OF PEDIATRICS DAY CARE

Minimizes parental separation

Uninterrupted feeding schedule/sleep patterns

Less risk of nosocomial infections

Convenience/Improved patient satisfaction

Availability of beds for complex ,needy patients

Reduced cost of hospitalization

PREMEDICATION

Aim

Minimizes effect of anesthetic agents

Prophylaxis's for aspiration

Decreases apprehension

Types of drugs-tranquilizers/opoid's anticholinergics /aspiration prophylaxis's

HOW TO MINIMIZE APPREHENSION IN CHILDREN

Limited waiting time before surgery

Attractive waiting room

Games for children

Limited separation time for children from families

Change hospital gown as close to the time of surgery

ANESTHETIC TECHNIQUES AND DRUGS

Evidence of successful anesthesia

Prompt post op presence of

Ambulation

Alertness

Analgesia

Alimentation

INDUCTION TECHNIQUE

Induction technique varies with

Facilities available

Anesthetists preference

Local custom

TECHNIQUES

Inhalational

Intravenous

Rectal

IM

Regional

INDUCTION

Inhalation and iv inductions are used in paediatric anaesthesia, and many factors influence the choice of induction method. Inhalation induction has traditionally been the preferred technique, since most children are needle-phobic. Inhalation induction is the choice in children with difficult venous access.

INTUBATION

Air leak 10-30cm H_2O

Consider different history e.g. downs syndrome

MONITORING

Use >2 monitoring devices

Precordial stethoscope

B/P

ECG

pulse oximetry

CHILDREN ARE LIABLE FOR HYPOXIA

Rationale

Low FRC

Increased metabolic rate

Increased oxygen consumption

PROCEDURES

Otolaryngology

Myringotomy and ventilating tubes, adenoidectomy , tonsillectomy, frenulectomy, branchial cleft cysts, endoscopic sinus surgery, examination under anaesthesia including some bronchoscopy

Ophthalmology

Examination under anaesthesia, strabismus repair, nasolacrimal duct probe, intraocular lens implantation, trabeculectomy

General pediatric surgery and urology

Herniorrhaphy and hydrocelectomy, orchiopexy, uncomplicated hypospadias, cystoscopy and cystoscopic surgery, circumcision, esophagoscopy, lumps

Gastroenterology

Endoscopy

Plastic surgery

Cleft lip and some cleft palate repair, placement of tissue expanders, scar revision, minor reconstructive procedures (otoplasty, septorhinoplasty, etc.)

Orthopaedics

Hardware removal, casting, percutaneous tenotomy, arthrograms Percutaneous pinnings and simple ORIF

Radiology

Imaging studies, radiation therapy

Dentistry

Extractions, restorations, examinations Nerve treatments, crowns, sealants, and fillings

EXCLUSION

Age and medical

Pre-term infant < 50 weeks post-conceptual age

Inadequately controlled systemic disease

Active viral or bacterial infection

Complex congenital heart disease

Un investigated cardiac murmur

Diabetes mellitus

Sickle cell disease

Sleep apnoea

SURGICAL AND ANESTHETIC

Inexperienced surgeon or anaesthetist

Prolonged procedure (> 1 h)

Risk of excessive peri-operative haemorrhage

Body cavity surgery

Difficult airway

Malignant hyperthermia

Postoperative pain unlikely to be controlled by simple analgesia

SOCIAL EXCLUSION

Lack of parental support

No telephone

Long journey home

Lack of transport facility

PROBLEMS IN PROVIDING DAYCARE IN CHILDREN

All children's are not medically uniform

Children with acute/chronic undiagnosed disease

Lost hospital records/recommendations

Inaccessible telephone, transport facilities

Illiterate parents, incapable of postoperative care

Absence from work/ extra visits (screening, postoperative follow up)

PREPARATION OF FAMILY AND CHILD

Family centered care

Preoperative teaching and parental presence

Premedication

ANESTHESIA FOR GERIATRICS DAY CARE SURGERY

Back ground

Geris –Greek >65years

5000 Americans reach this age every day

Significance

50% undergo surgery prior to death

CONSIDERATIONS

Presence of concomitant disease

Age related decrements in muti organ function

PHYSIOLOGIC CHANGES AND ANESTHETIC CONSIDERATIONS

1-Reduced autonomic end organ responsiveness

Anesthetic agents which disrupt autonomic end organ function and decrease catecholamine release or agents that produce ganglionic blockade can cause hypotension

BODY COMPOSITION

Cellular dehydration –by age of 80 years 8kgof skeletal muscle lost 12kg adipose gained TBW reduced to 80%increased peripheral compartment and decreased central compartment .

ELIMINATION OF DRUGS FROM PLASMA

Elimination of drugs is impaired by loss of hepatic mass or reduced renal perfusion

Increased duration of drug effect

Lipophilic molecule unchanged

Hydrophilic molecule (reduced distribution) increased plasma concentration if administered on kg/wt basis.

NEUROANATOMICAL BASIS FOR REDUCED DRUG NEED

Reduced myelination

Reduced specialized pain sensors

Reduced capacity for clearance (decreased hepatic and renal clearance

PREOPERATIVE EVALUATION

2-7days prior to surgery by three specialists

Adverse drug reaction (70%-10%)

CRITERIA'S FOR ADMISSION APPROPRIATENESS

Normal ---- Lab/V/S neurology /circulatory /respiratory /ECG/CXR and no active bleeding

PREMEDICATION

Not necessary (if a need use midazolam IV)

PROCEDURES

Not associated with major bleeding

Short duration

No major physiologic upset

TECHNIQUES OF ANESTHESIA

Regional /GA /monitored anesthesia care

RECOVERY

Longer to achieve street fitness

Rationale

Diminished metabolic rate

Decreased hepatic and renal clearance

INSTRUCTIONS &POST OPERATIVE FOLLOW UP

On how and where help can be contacted

This is mandatory to truly learn what is better or at least as good and not just less expensive

GENERAL CONSIDERATION FOR DAY CARE ANESTHESIA

Why day care surgery has become the fastest growing area in the health care delivery system?

ANSWER

Because of its ability to provide high quality and cost effective care

WHAT SHOULD BE DONE TO IMPROVE PRACTICE OF ANESTHESIA FOR DAY CARE IN ETHIOPIA ?

Set criteria's for selection of patients and operation procedures

Appropriate preoperative evaluation and laboratory tests

Rationale for premedication

Choice of anesthetic technique

Newer anesthetics and analgesia

Discharge criteria

PREOPERATIVE EVALUATION

Willingness of the patient

Proper selection of the patient

No age restriction

Duration 2-4 hours

ANESTHETIC TECHNIQUES AND DRUGS

Successful anesthesia for day care surgery is evidenced by prompt post op presence of

Alertness

Ambulation

Alimentation

analgesia

REGIONAL ANESTHESIA

advantages

Minimizes side effect of GA

Good post operative analgesia

Cost effective

DRAW BACKS OF REGIONAL

Time to perform block Vs time of onset is delayed

When procedure is short

When turn over time between cases is rapid

Prolonged motor blockade

SELECTION OF LOCAL ANESTHETIC DRUGS

Select short acting ones to avoid unwanted post op skeletal muscle weakness

INJECTABLE DRUGS

Influence greater than inhalational agents

Ciprofol, a new intravenous anaesthetic, is a gamma-aminobutyric acid (GABA) receptor agonist and is a novel 2,6-disubstituted phenol derivative like propofol with improved pharmacokinetic and pharmacodynamic characteristics. It has a rapid rate of onset and recovery in preclinical experiments.

ADVANTAGE OF TIVA

Administration of TIVA offers the following key advantages:

It does not require an anesthesia machine and scavenging equipment

It is associated with lower incidence of PONV

Avoids the risk of malignant hyperthermia

Provides rapid recovery without agitation.

DISADVANTAGE

TIVA has some limitations when compared with inhaled agents:

TIVA lacks the muscle relaxant effects of inhaled anaesthesia

In contrast to inhaled anaesthesia where the end-tidal concentrations can be used for prevention of recall, bispectral index (BIS) monitoring is recommended during TIVA. (The Bispectral Index™ (BIS™) monitoring system processes EEG information to provide a direct measurement of the patient's level of consciousness)

STANDARD TIVA

The standard TIVA protocol consists of administration of propofol with or without remifentanil. Propofol and remifentanil have synergistic effect .

DESIRABLE CHARACTERISTICS OF INJECTABLE DRUGS

Water soluble and stable in a solution

High therapeutic safety index

Non irritating after parenteral administration

No allergic reaction

Rapid and smooth onset of action

CONTINUED

No cardiopulmonary depression

Rapid degradation to inactive non toxic metabolites

Short elimination half time

Analgesia at sub anesthetic level

Rapid and smooth emergence with out side effects

INDUCTION AGENTS

Thiopentone 3-4mg /kg /methohexital1-1.5mg /kg/rapid recovery than thiopentone

.

Ethomidate-pian on injection/myoclonic movements/suppression of adrenocortical function (no steroid hormones)

Profofol-1.5-2mg/kg rapid onset prompt recovery than barbiturates impairment of psychic function in 30 minutes increased hypotension and depression of ventilation

PROFOFOL+LMA

MUSCLE RELAXANTS

Shot acting-suxamethonium(rapid onset&recovery)

Mivacurium-priming dose 0.03mg /kg 90 seconds to intubate(intubating dose -0.2-0.25mg/kg spontaneous recovery in 30 minutes shorter than vecuronium and atracurium .

Atracurium-0.25-0.3mg/kg

Vecuronium-0.04mg/kg-0.05mg/kg

OPIOIDS

Always use short acting ones

Alfentanyl 0.5-1mg

sufentanyl 10-15ug

fentanyl 3ug/kg or 50-100ug

BENZODIAZEPINES

Midazolam –fast acting and water soluble

INHALATIONAL AGENTS

Superior than Iv agents

Rapid elimination

More controllable

PREFERENCE OF INHALATIONAL AGENTS

With low blood gas solubility

Rapid onset

Prompt recovery

You can use (halothane/ sevoflurane/ enflurane NO2

INHALATIONAL AGENTS

Sevoflurane-rapid onset prompt recovery

Halothane –good for induction

Enflurane –extremely popular for day care surgery

N2Omain stay for day care surgery(60-70%+inhalational)popular.

FACTORS ASSOCIATED WITH POST OPERATIVE MORBIDITY

Lack of previous exposure

ETT(intubation draw backs)

Operation time no correlation with anesthesia

Female sex

ONE STUDY

65% after regional

26%after GA unfit for work next day

PACU MANAGEMENT

Manage most common problems

Pain management

Nausea and vomiting (relieved when pain is managed)

Discharge in less than 1.5Hrs

DELAY OF DISCHARGE

Excessive bleeding

Pain

Prolonged emergence

Dizziness

Intractable nausea and vomiting

Absence of responsible adults

TRIGGER DOT TEST

Fully awake oriented to time place and person

Stable Vs 30-60minutes

DISCHARGE CRITERIA AFTER GENERAL ANESTHESIA

Responsible adult

Stable Vs

No nausea and vomiting

Voided clear urine

Controlled pain

Good sensation and circulation in the operated area

CONTINUED

No evidence of bleeding

No signs of upper air way obstruction

Ability to recognize time and place

Ability to ambulate unassisted

Little or no evidence of dizziness after changing clothes or sitting for 10 minutes

DISCHARGE CRITERIA AFTER REGIONAL

Normal sensation

Ability to walk

Ability to urinate

INSTRUCTIONS

Don’t drive until tomorrow

About diet

How to manage myalgia

Defer important decisions and activities requiring special skill(96Hrs)

THANK YOU